

Chapter 05 Homework — Tests for irreducibility

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Problem 05.4. Suppose that K is a field and that f and g are relatively prime in $K[x]$. Show that $f - yg$ is irreducible in $K(y)[x]$.

Proof. Suppose $f - yg = ab$ with $a, b \in K(y)[x]$. We prove that a or b is a unit.

Since $f - yg \in K[y][x]$, Gauss's lemma (over the unique factorisation domain $K[y]$) gives a nonzero $c \in K(y)$ with $ca, c^{-1}b \in K[x][y]$. Scaling by c does not change whether a factor is a unit of $K(y)[x]$, so without loss of generality $a, b \in K[x][y]$.

In $K[x][y]$, $f - yg = f + (-g)y$ has degree 1 in y , and degrees in y add, so without loss of generality a has degree 0 and $b = b_0 + b_1y$, with $a, b_0, b_1 \in K[x]$. Then $ab = ab_0 + ab_1y$, which means $f = ab_0$ and $g = -ab_1$, so f and g have the common factor a . Since f and g are relatively prime, a must be a unit in $K[x]$. So it is a unit in $K(y)[x]$, since $K[x] \subseteq K(y)[x]$. $\square$

Problem 05.6. Suppose that R is an integral domain and that

$$f = f_0 + f_1x + \cdots + f_nx^n \in R[x]$$

has the property that $f_0, \dots, f_n$ are relatively prime. Suppose that p is a prime in R , and that $p \mid f_i$ for $1 \leq i \leq n$, while p does not divide f_0 and p^2 does not divide f_n . Show that f is irreducible in $R[x]$.

Proof. Reduce the coefficients of f modulo p . Since $p \mid f_i$ for $1 \leq i \leq n$, we get $\bar{f} = \bar{f}_0$, which is of degree zero, as $p \nmid f_0$.

Suppose $f = gh$ with $g, h \in R[x]$. Then $\bar{f} = \bar{g}\bar{h}$, and since (p) is prime, $R/(p)$ is an integral domain, so degrees add and both $\bar{g}$ and $\bar{h}$ must be of degree zero.

Case 1: g and h both of positive degree. Write $k = \deg g$ and $m = \deg h$, so $f_n = g_k h_m$. Since $\bar{g}$ and $\bar{h}$ are constants, the leading coefficients vanish mod p : $p \mid g_k$ and $p \mid h_m$. Hence $p^2 \mid f_n$, a contradiction.

Case 2: g or h of degree zero. Without loss of generality $\deg h = \deg f$ and $g(x) = c \in R$. Then $f_i = ch_i$ for all i , so $c \mid f_i$ for every i . Since the f_i are relatively prime, c must be a unit, and so f is irreducible in this case too. $\square$

Problem 05.7. Show that if p is a prime number then $x^n - p$ is irreducible in $\mathbb{Q}[x]$.

Proof. Write $x^n - p = a_0 + a_1x + \cdots + a_nx^n$, so $a_0 = -p$ and $a_n = 1$. Clearly $p \mid a_0$, $p^2 \nmid a_0$ and $p \nmid a_n$. For $1 \leq i < n$ we have $a_i = 0 = p \cdot 0$, so $p \mid a_i$. This is a direct example of Eisenstein's criterion, so $x^n - p$ is irreducible over $\mathbb{Q}$. $\square$

Problem 05.8. Let A denote the field of real numbers which are algebraic over $\mathbb{Q}$. Show that $[A : \mathbb{Q}] = \infty$.

Proof. Let $n \in \mathbb{N}$ be arbitrary. We will show $[A : \mathbb{Q}] > n$.

Consider $x^n - 2 \in \mathbb{Q}[x]$. Since it is irreducible in $\mathbb{Q}[x]$ by Eisenstein (05.7 with $p = 2$), $[\mathbb{Q}(\sqrt[n]{2}) : \mathbb{Q}] = n$.

As $\mathbb{Q}(\sqrt[n]{2}) \subsetneq A$, since for example $\sqrt[n+1]{2} \in A$ but $\sqrt[n+1]{2} \notin \mathbb{Q}(\sqrt[n]{2})$, clearly $[A : \mathbb{Q}] > [\mathbb{Q}(\sqrt[n]{2}) : \mathbb{Q}] = n$. Since n was arbitrary, $[A : \mathbb{Q}] = \infty$. $\square$

Problem 05.10. Show that $x^5 - 4x + 2$ and $x^4 - 4x + 2$ are irreducible over $\mathbb{Q}(i)$.

Proof. The quintic. Let α be a root of $x^5 - 4x + 2$. By Eisenstein at $p = 2$ this polynomial is irreducible in $\mathbb{Q}[x]$, so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 5$. Also $[\mathbb{Q}(i) : \mathbb{Q}] = 2$.

$[\mathbb{Q}(i, \alpha) : \mathbb{Q}(i)] \leq 5$, because α is a root of $x^5 - 4x + 2 \in \mathbb{Q}(i)[x]$. That polynomial may be reducible over $\mathbb{Q}(i)$. If it is, $[\mathbb{Q}(i, \alpha) : \mathbb{Q}(i)] < 5$; if it is not, $[\mathbb{Q}(i, \alpha) : \mathbb{Q}(i)] = 5$.

By the tower law applied to $\mathbb{Q} \subset \mathbb{Q}(i) \subset \mathbb{Q}(i, \alpha)$, the degree $[\mathbb{Q}(i, \alpha) : \mathbb{Q}]$ is at most $5 \cdot 2 = 10$. Applied to $\mathbb{Q} \subset \mathbb{Q}(\alpha) \subset \mathbb{Q}(i, \alpha)$ as well, both 2 and 5 must divide it, so it is 10. The only way to satisfy this is if $[\mathbb{Q}(i, \alpha) : \mathbb{Q}(i)] = 5$, so $x^5 - 4x + 2$ is irreducible in $\mathbb{Q}(i)[x]$, since it must be α 's minimal polynomial over $\mathbb{Q}(i)$.

The quartic. Clearly Eisenstein at $p = 2$ shows $x^4 - 4x + 2$ is irreducible in $\mathbb{Q}[x]$. Since $g(0) = 2 > 0$ and $g(1) = -1 < 0$, the intermediate value theorem gives a real root β of $g(x) = x^4 - 4x + 2$, and $[\mathbb{Q}(\beta) : \mathbb{Q}] = 4$. Note also $[\mathbb{Q}(i) : \mathbb{Q}] = 2$.

So $[\mathbb{Q}(i, \beta) : \mathbb{Q}] = 4$ or 8: it is divisible by 4 through $\mathbb{Q}(\beta)$, and (for the same argument as the quintic) $[\mathbb{Q}(i, \beta) : \mathbb{Q}(i)]$ is certainly not more than 4, so it is at most 8 through $\mathbb{Q}(i)$.

Suppose $[\mathbb{Q}(i, \beta) : \mathbb{Q}] = 4$. Then $[\mathbb{Q}(i, \beta) : \mathbb{Q}(\beta)] = 1$, meaning $i \in \mathbb{Q}(\beta)$. But this can't be, since $\beta \in \mathbb{R}$ gives $\mathbb{Q}(\beta) \subset \mathbb{R}$. So $[\mathbb{Q}(i, \beta) : \mathbb{Q}] = 8$, hence $[\mathbb{Q}(i, \beta) : \mathbb{Q}(i)] = 4$, which means $x^4 - 4x + 2$ is irreducible over $\mathbb{Q}(i)$, since it must be β 's minimal polynomial over $\mathbb{Q}(i)$. $\square$